

Lecture 16: Gaussian Elimination (continued)

Instructors:

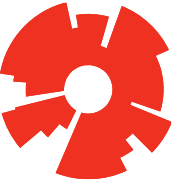
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[Based on slides for „Decoding the Matrix“]



WEEK





Agenda

Chapter 7

Lecture 16:

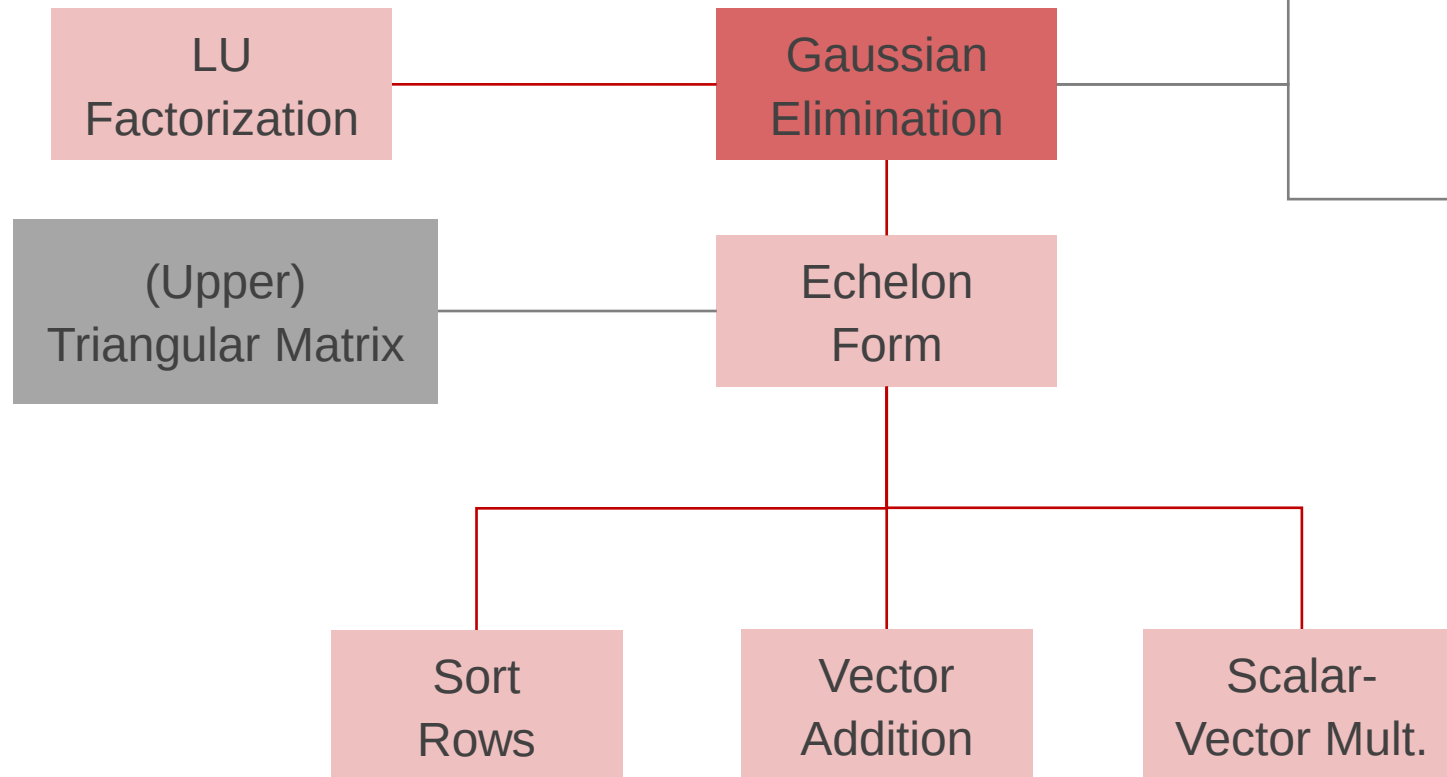
- Gaussian Elimination
- Gaussian Elimination - Solve Matrix Equation
- Gaussian Elimination - Basis for Null Space
- Gauss-Jordan Method
- Factoring Integers
- Factoring Integers - Naive Way
- Factoring Integers - Using Square Roots
- Greatest Common Divisor (GCD)



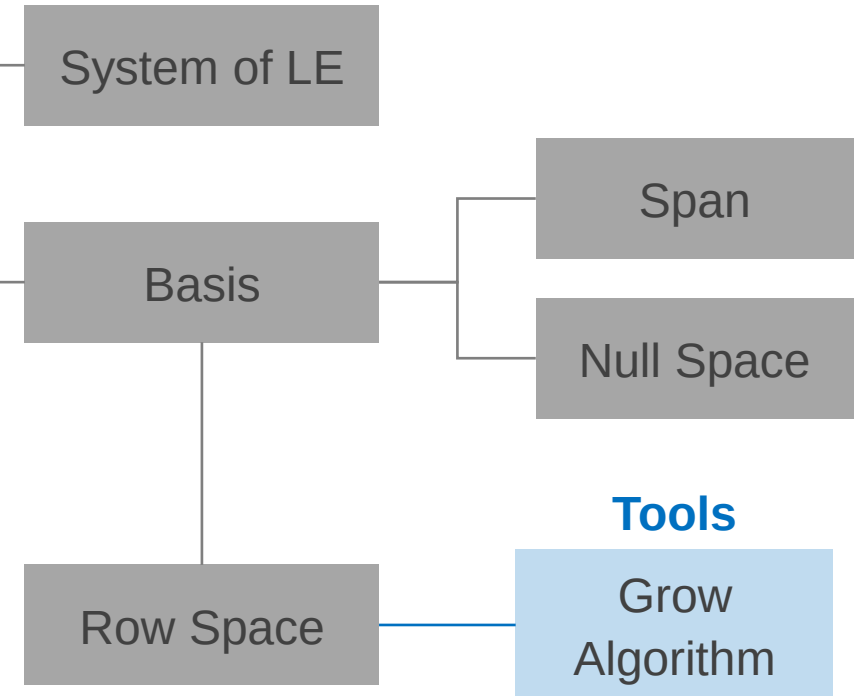


Lecture 15 - Recap

Concepts



Previous



Tools



Gaussian Elimination - Transformations

Key idea: keep track of transformations performed in putting matrix in echelon form.

Given matrix A , compute matrices M and U such that $MA = U$

- ▶ U is in echelon form
- ▶ M is invertible

$$\begin{bmatrix} M \\ \text{invertible} \end{bmatrix} \begin{bmatrix} A \end{bmatrix} = \begin{bmatrix} U \\ \text{Echelon Form} \end{bmatrix}$$

One matrix that does it all
(not separate matrices)



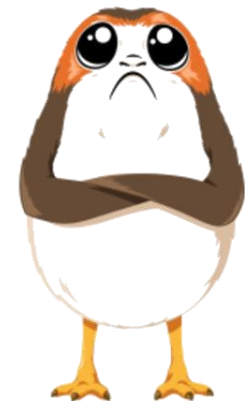
Gaussian Elimination - Example 1 (I)

Let's assume we have the following matrix A $\begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix}$ Let's bring it into Echelon Form!

Start

$$\begin{array}{c} \text{Identity Matrix} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ M \end{array} * \begin{array}{c} \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} \\ A \end{array} = \begin{array}{c} \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} \\ U \\ \text{(not Echelon form yet)} \end{array}$$

NOT IMPRESSED!





Nice!



Gaussian Elimination - Example 1 (II)

Start

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} \begin{matrix} Row_1 \\ Row_2 \\ Row_3 \end{matrix}$$

Iteration 1

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & 1 \\ 1 & 2.5 & 4 \end{bmatrix} \begin{matrix} Row_1 \\ Row_2 - 2 * Row_1 \\ Row_3 \end{matrix}$$

Select **first row (nonzero entry)** and use it to put the nonzero entry in the next row to **zero**

Iteration 2

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -0.5 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 2 & 3 \end{bmatrix} \begin{matrix} Row_1 \\ Row_2 \\ Row_3 - 0.5 * Row_1 \end{matrix}$$

Use **first row (nonzero entry)** to put nonzero entry in the last row to **zero**

Iteration 3

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3.5 & -2 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} Row_1 \\ Row_2 \\ Row_3 - 2 * Row_2 \end{matrix}$$

Use **second row (nonzero entry)** to put nonzero entry in the last row to **zero**



Gaussian Elimination - Example 1 (III)

After Iteration 3

$$\begin{matrix}
 \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3.5 & -2 & 1 \end{bmatrix} & * & \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} & = & \begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \\
 M & & A & & U \\
 & & \text{(unchanged)} & & \text{(Echelon form)}
 \end{matrix}$$



Remember (last lecture)

Lemma: If a matrix is in echelon form, the nonzero rows form a basis for the row space.

A different(?) question: What is a basis for $\text{Span}\{[2, 1, 2], [4, 3, 5], [1, 2.5, 4]\}$?

Answer: $\{[2, 1, 2], [0, 1, 1], [0, 0, 1]\}$

Bonus Insight: Rank of original matrix A is 3. It contains 3 rows and 3 linearly independent vectors form the basis for it → **linear independence**

(row rank = column rank)

I was here all along!





Gaussian Elimination - Recap: Uses

We just did an example of this!
Let's do an **example for GF(2)**



- ▶ *Finding a basis for the span of given vectors.* This additionally gives us an algorithm for rank and therefore for testing linear dependence.
- ▶ *Solving a matrix equation*, which is the same as *expressing a given vector as a linear combination of other given vectors*, which is the same as *solving a system of linear equations*
- ▶ *Finding a basis for the null space of a matrix*, which is the same as *finding a basis for the solution set of a homogeneous linear system*, which is also relevant to representing the solution set of a general linear system.
- ▶ Finding the **inverse matrix for a matrix** using an augmented matrix [**Gauss-Jordan Method**]



Gaussian Elimination - System of Equations (I)

Key idea: keep track of transformations performed in putting matrix in echelon form.

Given matrix A , compute matrices M and U such that $MA = U$

- ▶ U is in echelon form
- ▶ M is invertible

$$\begin{bmatrix} M \\ \text{invertible} \end{bmatrix} \begin{bmatrix} A \end{bmatrix} = \begin{bmatrix} U \\ \text{Echelon Form} \end{bmatrix}$$

To solve $A\mathbf{x} = \mathbf{b}$:

- ▶ Compute M and U so that $MA = U$
- ▶ Compute the matrix-vector product $M\mathbf{b}$, and solve $U\mathbf{x} = M\mathbf{b}$.

Using Echelon form matrix U is easier to solve than using A

Claim: This gives correct solution to $A\mathbf{x} = \mathbf{b}$





Gaussian Elimination - System of Equations (II)

Claim: This gives correct solution to $Ax = b$



Proof: Suppose $\mathbf{v}$ is a solution to $U\mathbf{x} = M\mathbf{b}$, so $U\mathbf{v} = M\mathbf{b}$

- ▶ Multiply both sides by M^{-1} : $M^{-1}(U\mathbf{v}) = M^{-1}M\mathbf{b}$
- ▶ Use associativity: $(M^{-1}U)\mathbf{v} = (M^{-1}M)\mathbf{b}$
- ▶ Cancel M^{-1} and M : $(M^{-1}U)\mathbf{v} = \mathbf{1}\mathbf{b}$
- ▶ Use $M^{-1}U = A$: $A\mathbf{v} = \mathbf{1}\mathbf{b} = \mathbf{b}$ [is the same as $U\mathbf{v} = M\mathbf{b}$]

How to solve $U\mathbf{x} = M\mathbf{b}$?

- ▶ If U is triangular, can solve using *back-substitution* (`triangular_solve`)
- ▶ In general, can use similar algorithm

Instead of $Ax = b$ we solve $Ux = Mb$
which is equivalent to $MAx = Mb$

U is basis for row space of A

M is matrix of transformations (to get A to U)



Gaussian Elimination - Solve Matrix Equation (I)

Let's assume we have the following set of general linear equations (right-hand side nonzero)

Linear Equations

$$\begin{aligned} 2x_1 + 1x_2 + 2x_3 &= 10 \\ 4x_1 + 3x_2 + 5x_3 &= 25 \\ 2x_1 + 2x_2 + 4x_3 &= 18 \end{aligned}$$

Matrix Vector Equation

$$\begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 2 & 2 & 4 \end{bmatrix} * \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 25 \\ 18 \end{bmatrix}$$

$$Ax = b$$

New Matrix Vector Equation

$$\begin{bmatrix} U \end{bmatrix} * \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} M \end{bmatrix} \begin{bmatrix} 10 \\ 25 \\ 18 \end{bmatrix}$$

$$Ux = Mb$$

$$\begin{aligned} a_1x_1 + a_2x_2 + a_3x_3 &= b_1 \\ a_4x_1 + a_5x_2 + a_6x_3 &= b_2 \\ a_7x_1 + a_8x_2 + a_9x_3 &= b_3 \end{aligned}$$

(Determine $\underbrace{MA}_{U} = U$ and then solve $Ux = Mb$)

**Gaussian
Elimination**



Gaussian Elimination - Solve Matrix Equation (II)

Matrix Vector Equation

$$\begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 2 & 2 & 4 \end{bmatrix} * \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 25 \\ 18 \end{bmatrix}$$

$$Ax = b$$

Starting Point for Gaussian Elimination

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 2 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 & | & 10 \\ 4 & 3 & 5 & | & 25 \\ 2 & 2 & 4 & | & 18 \end{bmatrix} \begin{matrix} Row_1 \\ Row_2 \\ Row_3 \end{matrix}$$

$$MA = U$$

Optional:
Include b here
(to directly get also Mb)

Nice! You are
changing the
equations as a whole!



Iteration
1

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 2 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 & | & 10 \\ 0 & 1 & 1 & | & 5 \\ 2 & 2 & 4 & | & 18 \end{bmatrix} \begin{matrix} Row_1 \\ Row_2 - 2 * Row_1 \\ Row_3 \end{matrix}$$

Iteration
2

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 2 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 & | & 10 \\ 0 & 1 & 1 & | & 5 \\ 0 & 1 & 2 & | & 8 \end{bmatrix} \begin{matrix} Row_1 \\ Row_2 \\ Row_3 - 1 * Row_1 \end{matrix}$$



Gaussian Elimination - Solve Matrix Equation (III)

Iteration 2

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 2 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 & | & 10 \\ 0 & 1 & 1 & | & 5 \\ 0 & 1 & 2 & | & 8 \end{bmatrix} \text{ Row}_3 - 1 * \text{Row}_1$$

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previous slide
Column 1 done

Iteration 3

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 2 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 & | & 10 \\ 0 & 1 & 1 & | & 5 \\ 0 & 0 & 1 & | & 3 \end{bmatrix} \text{ Row}_3 - 1 * \text{Row}_2$$

Column 2 done
... nothing else to do ...
Column 3 done too

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}}_M * \underbrace{\begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 2 & 2 & 4 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}}_U \underbrace{\begin{bmatrix} 10 \\ 5 \\ 3 \end{bmatrix}}_{Mb} \text{ Bonus (optional)}$$

Now let's use the result to
solve the new matrix-vector
equation



Gaussian Elimination - Solve Matrix Equation (IV)

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 2 & 2 & 4 \end{bmatrix} = \underbrace{\begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}}_U \underbrace{\begin{bmatrix} 10 \\ 5 \\ 3 \end{bmatrix}}_{Mb} \quad \text{Bonus (optional)}$$

$M \qquad A \qquad U \qquad Mb$

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 25 \\ 18 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \\ 3 \end{bmatrix}$$

$M \qquad b \qquad Mb$

If we did not adjust b during Gaussian Elimination, we could simply calculate Mb using M

Now let's use the result to solve the new matrix-vector equation $Ux = Mb$

Backward Substitution

$$\begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \\ 3 \end{bmatrix}$$

$U \qquad x \qquad Mb$

$$\begin{aligned} 2x_1 + 1x_2 + 2x_3 &= 10 \\ 1x_2 + 1x_3 &= 5 \\ 1x_3 &= 3 \end{aligned} \quad \rightarrow \quad \begin{aligned} 2x_1 + 1x_2 + 2(3) &= 10 \\ 1x_2 + 1(3) &= 5 \\ 1x_3 &= 3 \end{aligned} \quad \rightarrow \quad \begin{aligned} x_1 &= 1 \\ x_2 &= 2 \\ x_3 &= 3 \end{aligned} \quad \text{😊}$$

Bonus: U has three nonzero columns with linearly independent rows + is square $\rightarrow U$ is invertible $\rightarrow$ unique solution (for $Ux = Mb$)



Gaussian Elimination - Solve Matrix Equation (V)

New Matrix Vector Equation

$$Ux = Mb$$

$$\begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \\ 3 \end{bmatrix}$$

U x Mb

Matrix Vector Equation

$$Ax = b$$

$$\begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 2 & 2 & 4 \end{bmatrix} * \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 10 \\ 25 \\ 18 \end{bmatrix}$$

A x b

$$\begin{aligned} 2 * 1 + 1 * 2 + 2 * 3 &= 10 \\ 4 * 1 + 3 * 2 + 5 * 3 &= 25 \\ 2 * 1 + 2 * 2 + 4 * 3 &= 18 \end{aligned}$$

The solution for $Ux = Mb$ is the solution for $Ax = b$

[We proved this earlier, but it is nice to double-check our result]





Gaussian Elimination - Recap: Uses

- ✓ ▶ *Finding a basis for the span of given vectors.* This additionally gives us an algorithm for rank and therefore for testing linear dependence.
- ✓ ▶ *Solving a matrix equation*, which is the same as *expressing a given vector as a linear combination of other given vectors*, which is the same as *solving a system of linear equations*
- ▶ *Finding a basis for the null space of a matrix*, which is the same as *finding a basis for the solution set of a homogeneous linear system*, which is also relevant to representing the solution set of a general linear system.
- ▶ Finding the *inverse matrix for a matrix* using an augmented matrix **[Gauss-Jordan Method]**

We just did an example of **solving a matrix equation**! Let's next do an example for the **basis of a Null Space**





Gaussian Elimination - Basis for Null Space (I)

Definition: Null space of a matrix A is $\{x : \underline{A * x = 0}\}$. Written $\text{Null } A$

Remember
(Lecture 8)

Let's assume we have a matrix A and want to find the **Null Space** for A

Matrix Vector Equation

$$\begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 4 & 2 & 4 \end{bmatrix} * \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$Ax = 0$$

! Different
Example

x is Null Space for A

If we use Gaussian Elimination with A itself,
then we would end up [steps intentionally omitted]:

$$\begin{array}{c} \rightarrow \end{array} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 4 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \leftarrow \begin{array}{c} U \end{array}$$

$M \qquad A$

$$\begin{array}{c} [-2 \quad 0 \quad 1] \end{array} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 4 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$x \qquad A \qquad 0$



Gaussian Elimination - Basis for Null Space (II)

Matrix Vector Equation

$$\begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 4 & 2 & 4 \end{bmatrix} * \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$Ax = \mathbf{0}$$

$$Ax = \mathbf{0}$$

x is Null Space for A

$$\begin{bmatrix} -2 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 4 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$x^T A = \mathbf{0}^T$$

$$(x^T A)^T = A^T x$$

Let's use the transpose to
(hopefully) get it into the right form

$$\begin{bmatrix} 2 & 4 & 4 \\ 1 & 3 & 2 \\ 2 & 5 & 4 \end{bmatrix} * \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A^T x = \mathbf{0}$$

x is Null Space for A^T

Not the same ...



$$\begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 4 & 2 & 4 \end{bmatrix} * \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \\ -4 \end{bmatrix}$$

$$Ax \neq \mathbf{0}$$

NOT in Null Space for A



Gaussian Elimination - Basis for Null Space (III)

How to proceed to find the Null Space of A and not the Null Space of A^T ?

$$Ax = 0$$

x is Null Space for A

Matrix Vector Equation

$$\begin{bmatrix} A \end{bmatrix} * \begin{bmatrix} x \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

$$Ax = 0$$

=

$$\begin{bmatrix} x^T \end{bmatrix} * \begin{bmatrix} A^T \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

$$Ax = 0$$

$$\downarrow (Ax)^T = 0^T$$

$$x^T A^T = 0^T$$



$$x^T A^T = 0^T$$



$$\begin{bmatrix} M \end{bmatrix} * \begin{bmatrix} A^T \end{bmatrix} = \begin{bmatrix} U \end{bmatrix}$$

$$\text{row } m_i = x_i^T$$

$$A^T$$

$$U$$

Use rows in M
corresponding to
zero rows in U



Gaussian Elimination - Basis for Null Space (IV)

Step 1: Compute A^T

$$\begin{array}{ccc} \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 4 & 2 & 4 \end{bmatrix} & \longrightarrow & \begin{bmatrix} 2 & 4 & 4 \\ 1 & 3 & 2 \\ 2 & 5 & 4 \end{bmatrix} \\ A & & A^T \end{array}$$

Step 2: Perform Gaussian Elimination using A^T instead of A

Start

$$\begin{array}{ccc} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 4 & 4 \\ 1 & 3 & 2 \\ 2 & 5 & 4 \end{bmatrix} & = & \begin{bmatrix} 2 & 4 & 4 \\ 1 & 3 & 2 \\ 2 & 5 & 4 \end{bmatrix} \\ M & A^T & U \end{array}$$

Row₁
Row₂
Row₃

Iteration 1

$$\begin{array}{ccc} \begin{bmatrix} 1 & 0 & 0 \\ -0.5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 4 & 4 \\ 1 & 3 & 2 \\ 2 & 5 & 4 \end{bmatrix} & = & \begin{bmatrix} 2 & 4 & 4 \\ 0 & 1 & 0 \\ 2 & 5 & 4 \end{bmatrix} \\ & & \text{Row}_2 - 0.5 * \text{Row}_1 \end{array}$$



Gaussian Elimination - Basis for Null Space (V)

Iteration 1

$$\begin{bmatrix} 1 & 0 & 0 \\ -0.5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 4 & 4 \\ 1 & 3 & 2 \\ 2 & 5 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 4 \\ 0 & 1 & 0 \\ 2 & 5 & 4 \end{bmatrix} \text{Row}_2 - 0.5 * \text{Row}_1$$

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previous slide

Iteration 2

$$\begin{bmatrix} 1 & 0 & 0 \\ -0.5 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 4 & 4 \\ 1 & 3 & 2 \\ 2 & 5 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 4 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \text{Row}_3 - 1 * \text{Row}_1$$

Column 1 done

Iteration 3

$$\begin{bmatrix} 1 & 0 & 0 \\ -0.5 & 1 & 0 \\ -0.5 & -1 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 4 & 4 \\ 1 & 3 & 2 \\ 2 & 5 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{Row}_3 - 1 * \text{Row}_2$$

Column 2 done
... nothing else to do ...
Column 3 done too

$$\begin{bmatrix} 1 & 0 & 0 \\ -0.5 & 1 & 0 \\ -0.5 & -1 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 4 & 4 \\ 1 & 3 & 2 \\ 2 & 5 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

M A^T U



Gaussian Elimination - Basis for Null Space (VI)

Step 3: Take rows in M corresponding to zero rows in U

(If there were multiple zero rows, there would be multiple vectors in M that form the basis for the Null Space A and, thus, span it)

$$\begin{array}{c}
 \begin{bmatrix} 1 & 0 & 0 \\ -0.5 & 1 & 0 \\ -0.5 & -1 & 1 \end{bmatrix} \quad M \quad * \quad \begin{bmatrix} 2 & 4 & 4 \\ 1 & 3 & 2 \\ 2 & 5 & 4 \end{bmatrix} \quad A^T = \begin{bmatrix} 2 & 4 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad U \quad \text{Zero row} \\
 \downarrow \quad \downarrow \\
 \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 4 & 2 & 4 \end{bmatrix} * \begin{bmatrix} -0.5 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
 Ax = 0
 \end{array}$$

Step 4: Make concluding statement

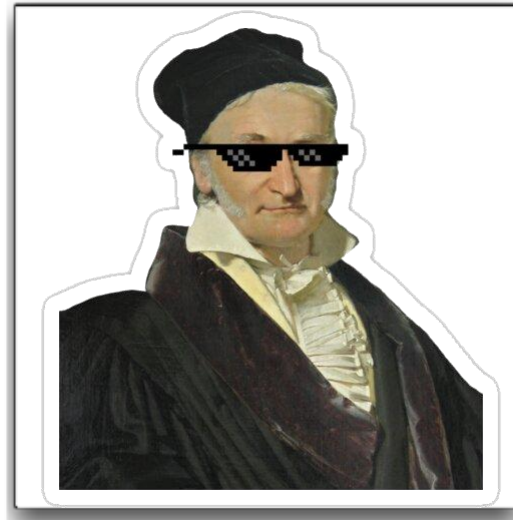
The vector $\begin{bmatrix} -0.5 \\ -1 \\ 1 \end{bmatrix}$ forms a basis for the Null Space of A

and the Null Space of A is, thus, $\text{Span} \left\{ \begin{bmatrix} -0.5 \\ -1 \\ 1 \end{bmatrix} \right\}$





Solve - Blackbox became less opaque



He was behind it
the entire time!

The modules `independence` and `solver` both use Gaussian elimination when working over $GF(2)$:

- ▶ The procedure `solve(A, b)` computes a matrix M such that MA is in echelon form, and uses M to try to find a solution.
- ▶ The procedure `rank(L)` converts to echelon form and counts the nonzero rows to find the rank of L .





Gaussian Elimination - Recap: Uses

- ✓ ▶ *Finding a basis for the span of given vectors.* This additionally gives us an algorithm for rank and therefore for testing linear dependence.
- ✓ ▶ *Solving a matrix equation*, which is the same as *expressing a given vector as a linear combination of other given vectors*, which is the same as *solving a system of linear equations*
- ✓ ▶ *Finding a basis for the null space of a matrix*, which is the same as *finding a basis for the solution set of a homogeneous linear system*, which is also relevant to representing the solution set of a general linear system.
- ▶ Finding the *inverse matrix for a matrix* using an augmented matrix [**Gauss-Jordan Method**]



Gauss-Jordan Method (I)

Not the same as Gaussian Elimination, but closely related



Gaussian Elimination:

- Goal: Find basis vectors for A / solving system of linear equations
- Result:
 1. Echelon Form matrix U (basis vectors)
 2. Matrix M with row operations

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3.5 & -2 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \text{Basis(Rows)}$$

Gauss Jordan:

- Goal: Find Matrix Inverse for A
- Result:
 1. Identity matrix U
 2. Matrix M is inverse

$$\begin{bmatrix} -0.25 & 0.5 & -0.5 \\ -5.5 & 3 & -1 \\ 3.5 & -2 & 1 \end{bmatrix} \text{Inverse} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{Identity}$$



Only works if
there is an inverse



Gauss-Jordan Method (II)

Gauss Jordan:

- (1) Additional steps are taken e.g., to ensure that the **leading scalar is a 1**
- (2) Additional row additions / subtractions done to **remove non-zero entries (off-diagonal)**

This could be done directly OR we could “continue” from the matrix containing the basis vectors (from Gaussian Elimination) until we have made it the identity matrix

$$\begin{array}{c} \text{Basis(Rows)} \\ \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3.5 & -2 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \end{array}
 \quad
 \begin{array}{c} \text{Inverse} \\ \begin{bmatrix} -0.25 & 0.5 & -0.5 \\ -5.5 & 3 & -1 \\ 3.5 & -2 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}
 \quad
 \begin{array}{c} \text{Identity} \end{array}$$



Only works if
there is an inverse



Gauss-Jordan Method (III)

$$\begin{array}{c} \text{Inverse} \\ \begin{bmatrix} -0.25 & 0.5 & -0.5 \\ -5.5 & 3 & -1 \\ 3.5 & -2 & 1 \end{bmatrix} \\ A^{-1} \end{array} * \begin{array}{c} \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} \\ A \end{array} = \begin{array}{c} \text{Identity} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ I \end{array}$$

Remember: We know that to solve $Ax = b$ we can use the inverse A^{-1} so that $x = bA^{-1}$. Thus, if the problem was:

$$\begin{array}{c} \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} \\ A \end{array} * \begin{array}{c} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ b \end{array} = \begin{array}{c} \begin{bmatrix} 6 \\ 15 \\ 10 \end{bmatrix} \\ b \end{array} \rightarrow \begin{array}{c} \begin{bmatrix} -0.25 & 0.5 & -0.5 \\ -5.5 & 3 & -1 \\ 3.5 & -2 & 1 \end{bmatrix} \\ A^{-1} \end{array} * \begin{array}{c} \begin{bmatrix} 6 \\ 15 \\ 10 \end{bmatrix} \\ b \end{array} = \begin{array}{c} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \\ x \end{array}$$

Wow! No System
of LEs needed!





Gauss-Jordan Method - Same Example (I)

Start $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} \begin{matrix} Row_1 \\ Row_2 \\ Row_3 \end{matrix}$

Iteration 1 $\begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0.5 & 1 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} Row_1/2$

The first value in the **first row** (nonzero entry) becomes a **one**

Iteration 2 $\begin{bmatrix} 0.5 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0.5 & 1 \\ 0 & 1 & 1 \\ 1 & 2.5 & 4 \end{bmatrix} Row_2 - 4Row_1$

Iteration 3 $\begin{bmatrix} 0.5 & 0 & 0 \\ -2 & 1 & 0 \\ -0.5 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0.5 & 1 \\ 0 & 1 & 1 \\ 0 & 2 & 3 \end{bmatrix} Row_3 - Row_1$

Same steps we would do for Gaussian Elimination for the matrix from the previous step



Gauss-Jordan Method - Same Example (II)

Iteration 3

$$\begin{bmatrix} 0.5 & 0 & 0 \\ -2 & 1 & 0 \\ -0.5 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0.5 & 1 \\ 0 & 1 & 1 \\ 0 & 2 & 3 \end{bmatrix} \text{ Row}_3 - \text{Row}_1$$

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Iteration 4

$$\begin{bmatrix} 0.5 & 0 & 0 \\ -2 & 1 & 0 \\ 3.5 & -2 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0.5 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \text{ Row}_3 - 2\text{Row}_2$$

In contrast to Gaussian Elimination, we keep going: triangular matrix, but not **identity matrix**

Iteration 5

$$\begin{bmatrix} 1.5 & -0.5 & 0 \\ -2 & 1 & 0 \\ 3.5 & -2 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0.5 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \text{ Row}_1 - 0.5\text{Row}_2$$

Iteration 6

$$\begin{bmatrix} -0.25 & 0.5 & -0.5 \\ -5.5 & 3 & -1 \\ 3.5 & -2 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 4 & 3 & 5 \\ 1 & 2.5 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} \text{Row}_1 - 0.5\text{Row}_3 \\ \text{Row}_2 - \text{Row}_3 \end{matrix}$$

Inverse

identity matrix & Inverse



Factoring Integers (I)

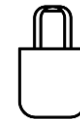
“A **prime number** [...] is a positive integer $p > 1$ that has no positive integer divisors other than 1 and p itself [...] meaning it is a number that cannot be factored.” - *Wolfram Mathworld*

Prime Factorization Theorem: For every integer $N \geq 1$, there is a unique bag of prime numbers whose product is N .

Example:

- ▶ 75 is the product of the elements in the bag $\{3, 5, 5\}$

$$75 = 3 * 5 * 5$$



- ▶ 126 is the product of the elements in the bag $\{2, 3, 3, 7\}$

$$126 = 2 * 3 * 3 * 7$$



- ▶ 23 is the product of the elements in the bag $\{23\}$

$$23 = 23$$



All the elements in a bag must be prime. If N is itself prime, the bag for N is just $\{N\}$.



Factoring Integers (II)

“The problem of distinguishing prime numbers from composite numbers and of **resolving the latter into their prime factors** is known to be one of the most important and useful in arithmetic. It has engaged the industry and wisdom of ancient and modern geometers to such an extent that it would be superfluous to discuss the problem at length. Further, the dignity of the science itself seems to require solution of a problem so elegant and so celebrated.”

Carl Friedrich Gauss, Disquisitiones Arithmeticae, 1801

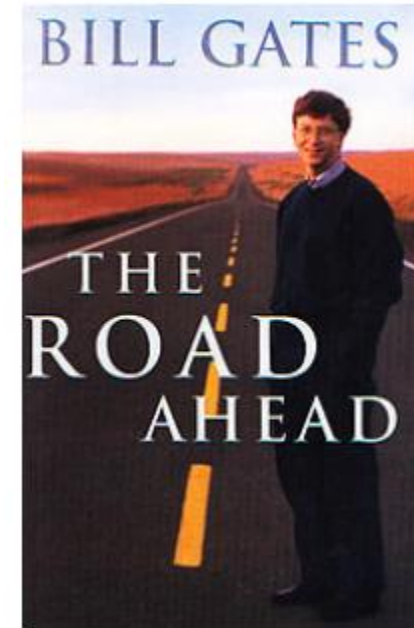




Factoring Integers (III)

“Because both the system’s privacy and the security of digital money depend on encryption, a breakthrough in mathematics or computer science that defeats the cryptographic system could be a disaster. The obvious mathematical breakthrough would be the development of an easy way to factor large composite numbers.”

(Bill Gates, The Road Ahead, 1995).



Secure communication with websites uses HTTPS (Secure HTTP) which is based on SSL (Secure Sockets Layer) which is based on the RSA (Rivest-Shamir-Adelman) cryptosystem which depends on the computational difficulty of factoring integers



Factoring Integers - Naive Way

```
def factor(N):
    for d in range(2, N-1):
        if N % d == 0: return d
```

[1 and N are trivial divisors for number N]

If d is a divisor of N then so is N/d .

$$\min\{d, N/d\} \leq \sqrt{N}$$

This shows that it suffices to search among $2, 3, \dots, \text{int}(\sqrt{N})$

```
def factor(N):
    for d in range(2, intsqrt(N)):
        if N % d == 0: return d
```

$N = 35$	$d = 2$	No (has remainder)
	$d = 3$	No (has remainder)
	$d = 4$	No (has remainder)
	$d = 5$	Yes

$N = 5$	$d = 2$	No (has remainder)
	$d = 3$	No (has remainder)
	$d = 4$	No (has remainder)

$N = 35$	$d = 2$	No (has remainder)
	$d = 3$	No (has remainder)
	$d = 4$	No (has remainder)
	$d = 5$	Yes

$N = 5$	$d = 2$	No (has remainder)
---------	---------	--------------------



Factoring Integers - Using Square Roots (I)

Find integers a and b such that $a^2 - b^2 = N$

for then $(a - b)(a + b) = N$ so $a - b$ and $a + b$ are divisors (ideally nontrivial)
Not 1 or N

How to find such integers? Naive approach using square roots

1. Choose integer a slightly more than $\sqrt{N}$
2. Check if $\sqrt{a^2 - N}$ is an integer.
3. If so, let $b = \sqrt{a^2 - N}$ Success! Now $a - b$ is a divisor of N
4. If not, repeat with another value for a



Factoring Integers - Using Square Roots (II)

1. Choose integer a slightly more than $\sqrt{N}$
2. Check if $\sqrt{a^2 - N}$ is an integer.
3. If so, let $b = \sqrt{a^2 - N}$ Success! Now $a - b$ is a divisor of N
4. If not, repeat with another value for a

Find

$$(a - b)(a + b) = N$$

Example: $N = 77$

1. $\sqrt{77} \approx 8.78 \rightarrow a = 9$
2. $b = \sqrt{(a^2 - N)} = \sqrt{(81 - 77)} = 2$ ✓
3. Yes $\rightarrow (a - b) = 7$ is a divisor of N
4. -

Example: $N = 943$

1. $\sqrt{943} \approx 30.71 \rightarrow a = 31$
2. $b = \sqrt{(a^2 - N)} = \sqrt{(961 - 943)} \approx 4.24$ ✗
4. No $\rightarrow a = a + 1 = 32$
 $b = \sqrt{(a^2 - N)} = 9$ ✓
 Yes $\rightarrow (a - b) = 23$ is a divisor of N



Factoring Integers - Using Square Roots (III)

$$a^2 - b^2 = N$$

$$(a - b)(a + b) = N$$

Example: $N = 77$ $a = 9, b = 2$

$$9^2 - 2^2 = 77$$

$$(9 - 2) * (9 + 2) = 77$$

$$7 * 11 = 77$$

Prime Factors

Example: $N = 943$ $a = 32, b = 9$

$$32^2 - 9^2 = 943$$

$$(32 - 9) * (32 + 9) = 943$$

$$23 * 41 = 943$$

Prime Factors



**Works, but slow
in practice**



Factoring Integers - GCD (I)

For large N , it takes too long to find a good integer a . 😞

We will show how **linear algebra** 😊 helps us synthesize a good integer a .

Let's first understand the subroutine of the greatest common divisor (gcd)

`gcd(m,n)` return the greatest common divisor of positive integers m and n . This algorithm is attributed to Euclid, and it is very fast. Here's the code:

```
def gcd(x,y): return x if y == 0 else gcd(y, x % y)
```

Until the second entry is zero, replace the second entry by the **remainder** of the division

At some point the **remainder** of the division will be zero $\rightarrow$ x will be the gcd

One of my great ideas from about 2300 years ago





Factoring Integers - GCD (II)

$\text{gcd}(m, n)$ return the greatest common divisor of positive integers m and n . This algorithm is attributed to Euclid, and it is very fast. Here's the code:

```
def gcd(x,y): return x if y == 0 else gcd(y, x % y)
```

What is the largest divisor for 252 and 105?

$\text{gcd}(252, 105) \rightarrow y = 105 \neq 0$

$$\frac{252}{105} = 2; \text{remainder} = 42$$

$\text{gcd}(105, 42) \rightarrow y = 42 \neq 0$

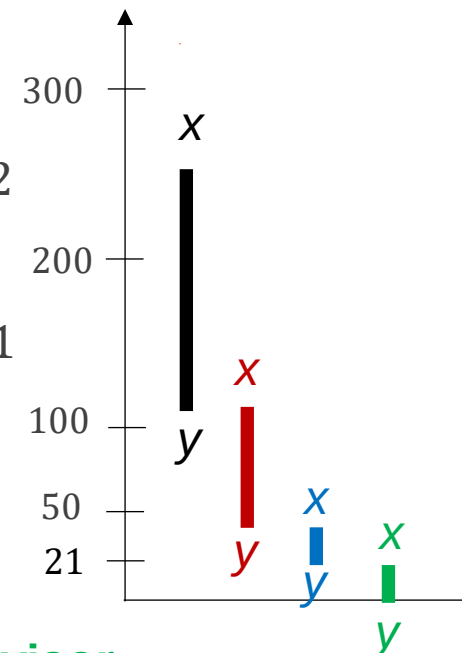
$$\frac{105}{42} = 2; \text{remainder} = 21$$

$\text{gcd}(42, 21) \rightarrow y = 21 \neq 0$

$$\frac{42}{21} = 2; \text{remainder} = 0$$

$\text{gcd}(21, 0) \rightarrow y = 0$ ✓

21 is greatest common divisor



Heureka! ... ah, wait, that is not my signature statement





Exercise: Gaussian Elimination

You are given the following matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

- 1) Using Gaussian Elimination, what are the basis vectors for the row space of A ? Show work!
- 2) Using Gaussian Elimination, what are the basis vectors for the Null A ? Show work!
- 3) Could we use the Gauss-Jordan Method to find A^{-1} ? How is this linked to the linear function invertibility theorem? Explain.



Solution

Exercise: Gaussian Elimination

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

a)

Start $\begin{matrix} M & A & U \end{matrix}$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Iteration 1

$$\begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{bmatrix} \text{ Row}_2 - 4\text{Row}_1$$

Iteration 2

$$\begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ -7 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix} \text{ Row}_3 - 7\text{Row}_1$$

Iteration 3

$$\begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix} \text{ Row}_3 - 2\text{Row}_2$$

(First Column done)

(Second Column done)
(No columns remaining)

There are two basis vectors for A and they are $[1, 2, 3], [0, -3, -6]$



Solution

Exercise: Gaussian Elimination

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

b) To obtain the basis vectors for Null A we need to perform Gaussian Elimination not for A but for A^T

$$\text{Start} \quad \begin{matrix} M & A^T & U \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} \end{matrix}$$

$$\text{Iteration 1} \quad \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 4 & 7 \\ 0 & -3 & -6 \\ 3 & 6 & 9 \end{bmatrix} \quad \text{Row}_2 - 2\text{Row}_1$$

$$\text{Iteration 2} \quad \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 4 & 7 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix} \quad \text{Row}_3 - 3\text{Row}_1$$

$$\text{Iteration 3} \quad \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 4 & 7 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{Row}_3 - 2\text{Row}_2$$

Gaussian Elimination
completed



Solution

Exercise: Gaussian Elimination

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$\begin{matrix} M & & A^T & & U \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 4 & 7 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

There is a single zero row vector in U so the corresponding row vector in M is the basis vector for $\text{Null } A$. Thus, the **basis vector(s) for $\text{Null } A$ is $[1, -2, 1]$.**

- c) **Linear-Function Invertibility Theorem:** Let $f : \mathcal{V} \longrightarrow \mathcal{W}$ be a linear function. Then f is invertible iff $\dim \text{Ker } f = 0$ and $\dim \mathcal{V} = \dim \mathcal{W}$.

For a linear function (e.g., $f(x) = Ax$) to be invertible, the kernel of the function needs to only contain the zero vector (dimension is zero). This is the same as $\text{Null } A = 0$. However, $\text{Null } A = \text{Span}([1, -2, 1])$, which means it has dimension 1. Thus, neither the linear function nor the corresponding matrix A are invertible.

Since there is no inverse A^{-1} , we cannot use Gauss-Jordan Method to find it (it will fail to construct the identity & inverse).



Some Key Take-aways (Lecture 16)

For **Gaussian Elimination**, we want to keep track of the transformations that we are performing to change a matrix A to the Echelon form U so that $MA = U$

Gaussian Elimination can be used for (1) finding a basis for the span of vectors, (2) solving a matrix equation $Ax = b$ and (3) finding a basis for the Null Space of a matrix

We had examples
for all three

To (2) **solve matrix equations**, we can solve $Ux = Mb$ instead of solving $Ax = b$ directly, by finding the echelon form of A via $MA = U$

To (3) **find the basis for the Null Space** of A , we solve $MA^T = U$ and the rows of M that correspond to zero rows in the Echelon form matrix U , form a basis for $\text{Null } A$

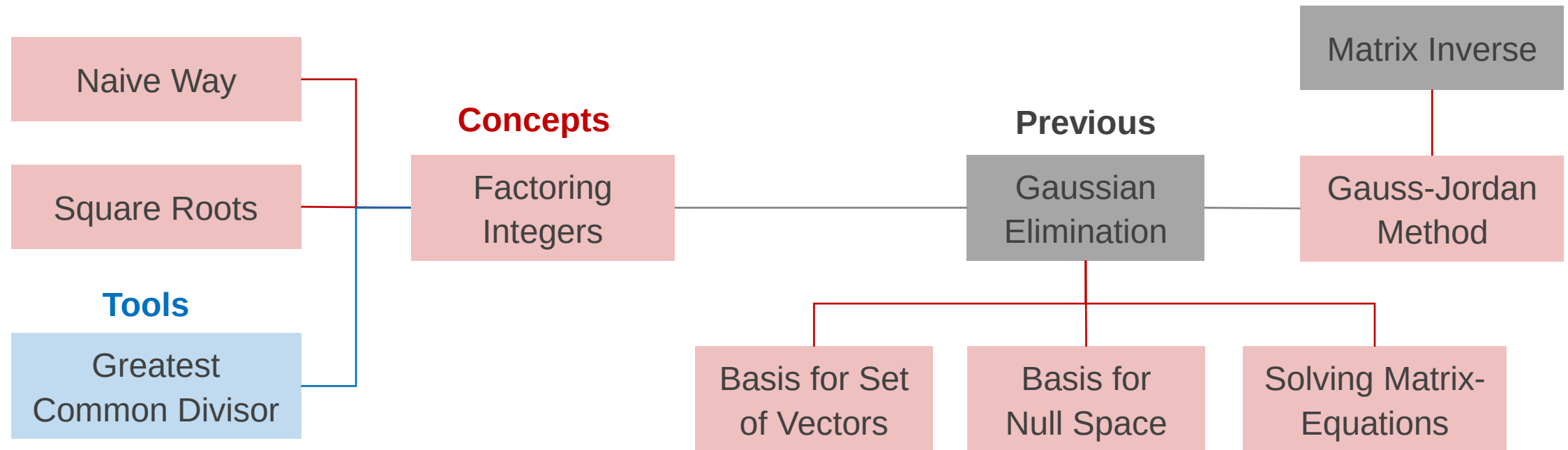
Factoring Integers concerns finding a bag of prime numbers whose product is a composite number N (integer)

Using the algorithm for **greatest common divisor (gcd)** can support in factoring integers but also $\text{GF}(2)$ and the idea of a perfect square can be used

Incl. Gaussian
Elimination



Lecture 16 - Summary





Thank You For Your Attention!



**Accomplishment unlocked:
Lecture 16 Completed**



APPENDIX



Gaussian Elimination - Example GF(2) (I)

Start

$$\begin{matrix}
 \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} & * & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} & = & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 M & & A & & U \text{ (not Echelon form yet)}
 \end{matrix}$$

Row₁
Row₂
Row₃
Row₄
Row₅

Iteration 1

$$\begin{matrix}
 \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} & * & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} & = & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 & & & & \text{Row}_2 + \text{Row}_1
 \end{matrix}$$

Select **first row (nonzero entry)**
and use it ensure all entries in
the first column in all other rows
are zero

Iteration 2

$$\begin{matrix}
 \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} & * & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} & = & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 & & & & \text{Row}_4 + \text{Row}_1
 \end{matrix}$$

Column 1 done



Gaussian Elimination - Example GF(2) (II)

Iteration 2

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{Row}_4 + \text{Row}_1$$

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Column 1 done

Iteration 3

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{Row}_3 + \text{Row}_2$$

Select **second row (nonzero entry)** and use it ensure all entries in the second column in all other rows are zero

Iteration 4

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{Row}_4 + \text{Row}_2$$

Column 2 (and 3) done



Gaussian Elimination - Example GF(2) (III)

Iteration 4

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{Row}_4 + \text{Row}_2$$

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Column 2 (and 3) done

Iteration 5

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{Row}_4 + \text{Row}_3$$

Select **third row (nonzero entry)** and use it ensure all entries in the fourth column in all other rows are zero

Iteration 6

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{Row}_5 + \text{Row}_3$$

End - first three rows are basis for row space of A



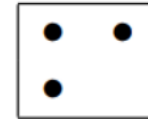
Gaussian Elimination for other Problems

So far:

- ▶ we know how to use Gaussian elimination to transform a matrix into echelon form;
- ▶ nonzero rows form a basis for row space of original matrix

We can do other things with Gaussian elimination:

- ▶ Solve linear systems (used in e.g. *Lights Out*)
- ▶ Find vectors in null space (used in e.g. integer factoring)



Key idea: keep track of transformations performed in putting matrix in echelon form.



Transformation to Echelon Form (I)

```
col_label_list = sorted(rowlist[0].D, key=str)
new_rowlist = []
rows_left = set(range(len(rowlist)))
```

Pre-processing

```
for c in col_label_list:
    rows_with_nonzero = [r for r in rows_left if rowlist[r][c] != 0]
    if rows_with_nonzero != []:
        pivot = rows_with_nonzero[0]
        rows_left.remove(pivot)
        new_rowlist.append(rowlist[pivot])
        for r in rows_with_nonzero[1:]:
            multiplier = rowlist[r][c]/rowlist[pivot][c]
            rowlist[r] -= multiplier * rowlist[pivot]
```

Take first row with nonzero value for current column

For other rows that also have a nonzero value in that column, adjust row so value is zero



Transformation to Echelon Form (II)

- ▶ Initialize rowlist to be list of rows of A
- ▶ Initialize $M_rowlist$ to be list of rows of identity matrix

```
for c in sorted(col_labels, key=str):
    rows_with_nonzero = [r for r in rows_left if rowlist[r][c] != 0]
    if rows_with_nonzero != []:
        pivot = rows_with_nonzero[0]
        rows_left.remove(pivot)
        new_M_rowlist.append(M_rowlist[pivot])
        for r in rows_with_nonzero[1:]:
            multiplier = rowlist[r][c]/rowlist[pivot][c]
            rowlist[r] -= multiplier*rowlist[pivot]
            M_rowlist[r] -= multiplier*M_rowlist[pivot]

        for r in rows_left: new_M_rowlist.append(M_rowlist[r])
```

Finally, return matrix M formed from $M_rowlist$



Factoring Integers - Using GCD (I)

For large N , it takes too long to find a good integer a . ☹️

We will show how **linear algebra** 😊 helps us synthesize a good integer a .

Find a and b such that

$$a^2 - b^2 = kN$$

for some integer k . Then

Now

$$(a - b)(a + b) = kN$$

Before

$$(a - b)(a + b) = N$$



Factoring Integers - Using GCD (II)

Now

$$(a - b)(a + b) = kN$$

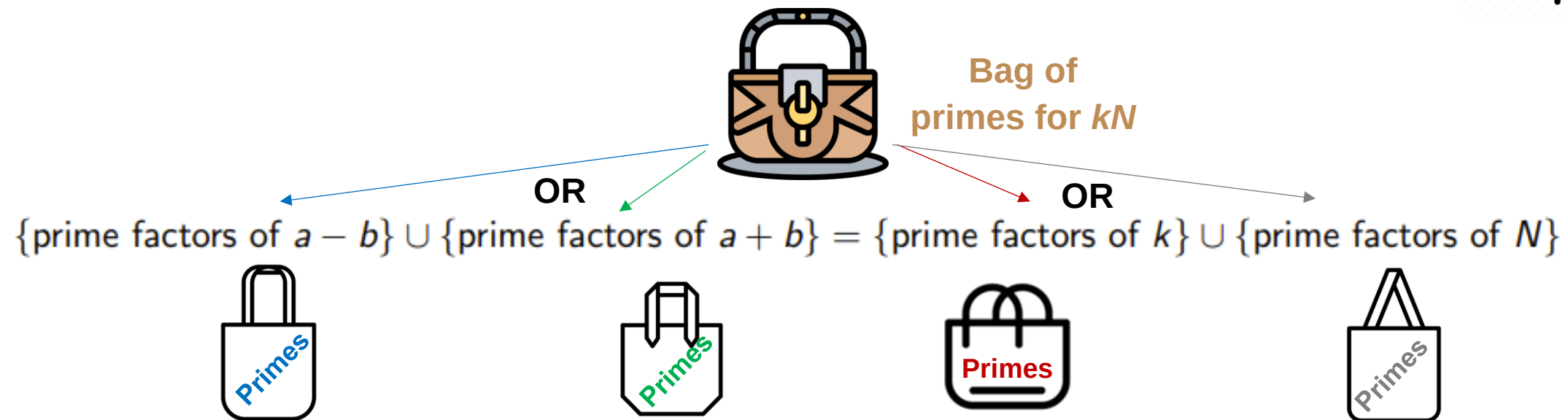
Before

$$(a - b)(a + b) = N$$

How is this easier? It does not look easier!



The left-hand side does not need to be equal to N anymore, it just needs to be a multiple of N
 (→ we lowered our standard / requirement)





Factoring Integers - Using GCD (III)

$$\{\text{prime factors of } a - b\} \cup \{\text{prime factors of } a + b\} = \{\text{prime factors of } k\} \cup \{\text{prime factors of } N\}$$



$\{q, p\}$

Same bag

IF $q, p \in (a - b)$ OR $q, p \in (a + b)$

THEN $\gcd(a - b, N)$ will only find **trivial** divisor 🙄

Different bag

IF $(p \in (a - b) \text{ AND } q \in (a + b))$ OR $(q \in (a - b) \text{ AND } p \in (a + b))$

THEN $\gcd(a - b, N)$ will find a **nontrivial** divisor 😊

I'll assist for the next examples



Omnipotent Owl



Factoring Integers - Using GCD (IV)

[$kN = 60'060$]



Let's assume we selected a and b as 272 and 118 [Example 1]

$$(a - b) = 154$$



$\{2, 7, 11\}$

$$(a + b) = 390$$



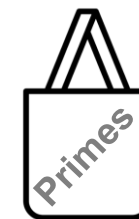
$\{2, 3, 5, 13\}$

$$k = 780$$



$\{2, 2, 3, 5, 13\}$

$$N = 77$$



$\{7, 11\}$

Same bag

$$\begin{aligned} \gcd(a - b, N) &= \gcd(154, 77) \longrightarrow y = 77 \neq 0 & \frac{154}{77} &= 2; \text{remainder} = 0 \\ \gcd(77, 0) &\longrightarrow y = 0 \quad \checkmark \end{aligned}$$

77 is greatest common divisor = N





Factoring Integers - Using GCD (V)

[$kN = 60'060$]



Let's assume we selected a and b as 328 and 218 [Example 2]

$$(a - b) = 110$$



$\{2, 5, 11\}$

$$(a + b) = 546$$



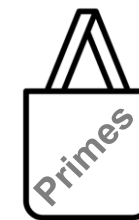
$\{2, 3, 7, 13\}$

$$k = 780$$

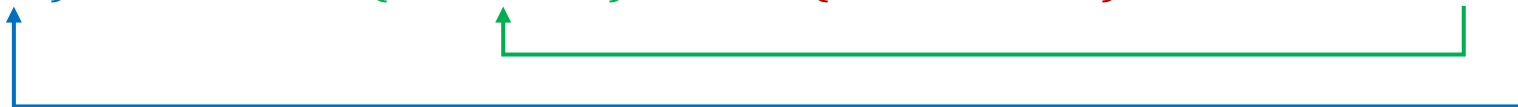


$\{2, 2, 3, 5, 13\}$

$$N = 77$$



$\{7, 11\}$



Different bag

$$\gcd(a - b, N) = \gcd(110, 77) \longrightarrow y = 77 \neq 0$$

$$\gcd(77, 33) \longrightarrow y = 33 \neq 0$$

$$\gcd(33, 11) \longrightarrow y = 11 \neq 0$$

$$\gcd(11, 0) \longrightarrow y = 0 \quad \checkmark$$

$$\frac{110}{77} = 1; \text{remainder} = 33$$

$$\frac{77}{33} = 2; \text{remainder} = 11$$

$$\frac{33}{11} = 3; \text{remainder} = 0$$

11 is greatest common divisor



Repeat same with $(a+b)=546 \rightarrow \gcd = 7$



Factoring Integers - Finding a and b (I)

Omnipotent Owl



That worked well, but remember, I helped you by telling you what a and b is. Can you find good ones yourself?

Find a and b so that $a^2 - b^2 = kN$

Idea: Start by finding the first thousand prime numbers $p_1, \dots, p_{1000}$. “*primeset*”

- 1 Iteration
- ▶ Choose a
 - ▶ Compute $a^2 - N$.
 - ▶ See if $a^2 - N$ can be factored using only $p_1, \dots, p_{1000}$
 - ▶ If not, throw it away.
 - ▶ If so, record a and the factorization of $a^2 - N$

Repeat a thousand and one times



Factoring Integers - Finding a and b (II)

Example: N is given as 2419

- ▶ Choose a
- ▶ Compute $a^2 - N$.
- ▶ See if $a^2 - N$ can be factored using only $p_1, \dots, p_{1000}$
- ▶ If not, throw it away.
- ▶ If so, record a and the factorization of $a^2 - N$

a	$a * a - N$	factorization
51	182	$2 \cdot 7 \cdot 13$
52	285	$3 \cdot 5 \cdot 19$
53	390	$2 \cdot 3 \cdot 5 \cdot 13$
58	945	$3^3 \cdot 5 \cdot 7$
61	1302	$2 \cdot 3 \cdot 7 \cdot 31$
62	1425	$3 \cdot 5^2 \cdot 19$
63	1550	$2 \cdot 5^2 \cdot 31$
67	2070	$2 \cdot 3^2 \cdot 5 \cdot 23$
68	2205	$3^2 \cdot 5 \cdot 7^2$
71	2622	$2 \cdot 3 \cdot 19 \cdot 23$

Remark: Course book author only did it until p_{32} and this is why some values of a were “thrown away” that can be factorized with e.g., values up to p_{1000}



Factoring Integers - Finding a and b (III)

a	$a * a - N$	factorization
51	182	$2 \cdot 7 \cdot 13$
52	285	$3 \cdot 5 \cdot 19$
53	390	$2 \cdot 3 \cdot 5 \cdot 13$
58	945	$3^3 \cdot 5 \cdot 7$
61	1302	$2 \cdot 3 \cdot 7 \cdot 31$
62	1425	$3 \cdot 5^2 \cdot 19$
63	1550	$2 \cdot 5^2 \cdot 31$
67	2070	$2 \cdot 3^2 \cdot 5 \cdot 23$
68	2205	$3^2 \cdot 5 \cdot 7^2$
71	2622	$2 \cdot 3 \cdot 19 \cdot 23$

Now we want to find a subset $\{a_1, \dots, a_k\}$ such that $(a_1^2 - N) \cdots (a_k^2 - N)$ is a perfect square.

Perfect square: an integer whose square-root is an integer (e.g., 9, 25 or 36)

Combine $a_1 = 52, a_2 = 67, a_3 = 71$



$$(a_1^2 - N)(a_2^2 - N)(a_3^2 - N) =$$

$$(3 \cdot 5 \cdot 19)(2 \cdot 3^2 \cdot 5 \cdot 23)(2 \cdot 3 \cdot 19 \cdot 23)$$

$$= 2^2 \cdot 3^4 \cdot 5^2 \cdot 19^2 \cdot 23^2 = (2 \cdot 3^2 \cdot 5 \cdot 19 \cdot 23)^2$$

Perfect Square

How to find a subset that works?



Factoring Integers - Finding a and b (IV)

Perfect square: an integer whose square-root is an integer (e.g., 9, 25 or 36)

Omnipotent Owl



I helped you again by telling which factorizations to use – next time you're on your own!

Combine $a_1 = 52$, $a_2 = 67$, $a_3 = 71$

$$\begin{aligned} (a_1^2 - N)(a_2^2 - N)(a_3^2 - N) &= \\ (3 \cdot 5 \cdot 19)(2 \cdot 3^2 \cdot 5 \cdot 23)(2 \cdot 3 \cdot 19 \cdot 23) &= \\ = 2^2 \cdot 3^4 \cdot 5^2 \cdot 19^2 \cdot 23^2 &= (2 \cdot 3^2 \cdot 5 \cdot 19 \cdot 23)^2 \\ &\text{Perfect Square} \\ &\text{(an integer squared)} \end{aligned}$$

Important Observation: Before we pulled out the power 2, all exponents were even (not odd)

So it matters that for each prime factor, the sum of the exponents is even and not odd (it does not matter which even number)





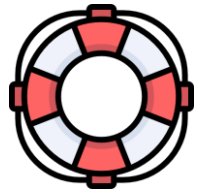
Factoring Integers - Finding a and b (V)

Represent each factorization as a vector over $GF(2)$:

Represent $p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$ by $\{p_1 : (a_1 \% 2), p_2 : (a_2 \% 2) \dots, p_k : (a_k \% 2)\}$

If even number, then **zero**, if odd number, then **one**

Linear algebra to
the rescue!



Let A = matrix whose rows are these vectors.

A subset of factorizations whose product is a perfect square = a subset of A 's rows whose sum is the zero vector (= all exponents in the product are even $\rightarrow$ needed for a perfect square)

Therefore need to find a nonzero vector in $\{\mathbf{u} : \mathbf{u} * A = \mathbf{0}\} \rightarrow$ Gaussian Elimination to find a basis for the Null Space of A^T
If number of rows $>$ rank of matrix then there exists such a nonzero vector.

I guess I'm not needed anymore! Goodbye Human!





Factoring Integers - Finding a and b (VI)

If even number, then **zero**, if odd number, then **one** - *[Example in blue below]*

a	a * a - N	factorization	vector.f
51	182	2 · 7 · 13	{2 : one, 13 : one, 7 : one}
52	285	3 · 5 · 19	{19 : one, 3 : one, 5 : one}
53	390	2 · 3 · 5 · 13	{2 : one, 3 : one, 5 : one, 13 : one}
58	945	3 ³ · 5 · 7	{3 : one, 5 : one, 7 : one}
61	1302	2 · 3 · 7 · 31	{31 : one, 2 : one, 3 : one, 7 : one}
62	1425	3 · 5 ² · 19	{19 : one, 3 : one, 5 : 0}
63	1550	2 · 5 ² · 31	{2 : one, 5 : 0, 31 : one}
67	2070	2 · 3 ² · 5 · 23	{2 : one, 3 : 0, 5 : one, 23 : one}
68	2205	3 ² · 5 · 7 ²	{3 : 0, 5 : one, 7 : 0}
71	2622	2 · 3 · 19 · 23	{19 : one, 2 : one, 3 : one, 23 : one}

2: 1 + 1 = 0

3: 1 + 0 + 1 = 0

5: 1 + 1 = 0

19: 1 + 1 = 0

23: 1 + 1 = 0



Zeros indicate that for all prime numbers the exponent is even
→ perfect square for the product of these factorizations